

PURCHASE-TO-PAY PROCESS MINING & EXCEPTION INTELLIGENCE

Business Case & Business Requirements Document

Data Analyst + Business Analyst Portfolio Project

Document Control	Details
Source dataset	BPI Challenge 2019 event log converted to CSV
Dataset scale	1,595,923 events 251,734 P2P cases 42 activities
Document version	1.1 - Recruiter-ready baseline
Purpose	Define the business problem, scope, requirements, KPIs, and analytical boundaries

Portfolio note: The source event log is anonymized. Stakeholder roles and the management scenario are portfolio constructs aligned to the data and are not claims about the source organization.

1. Executive Summary

This project analyzes a large Purchase-to-Pay (P2P) event log to understand how transactions are actually executed, where operational exceptions and rework occur, which process paths experience long cycle times, and which cases or process segments should be prioritized for investigation. The Business Analysis layer defines the business problem, stakeholders, scope, requirements, and success measures; the Data Analytics layer translates those needs into event-level and case-level analytical models, SQL analysis, Python exploration, Excel review, and Power BI reporting.

Dataset Baseline	Validated Value
Event rows	1,595,923
Columns	21
Unique P2P cases	251,734
Purchasing documents	76,349
Vendors	1,975
Activities	42
Companies	4

2. Business Problem

A multinational organization processes a high volume of P2P transactions across vendors, purchasing processes, spend categories, and operational resources. Although many transactions follow common purchasing and invoice-processing paths, others contain purchase-order modifications, approval changes, goods-receipt reversals, invoice reversals, payment-block handling, repeated activities, and unusual process sequences. Management lacks a consolidated analytical view of how transactions actually move through the P2P process, where operational rework and processing delays occur, and which cases or process segments contribute disproportionately to process complexity.

The business therefore requires a data-driven P2P process-intelligence solution capable of reconstructing actual transaction flows, measuring process performance, identifying exception and rework patterns, and prioritizing cases and process areas requiring investigation and improvement.

3. Problem Statement

The current P2P event data contains enough information to reconstruct case-level process paths, but it is not yet organized into a management-ready analytical model. As a result, Procurement, Accounts Payable, Finance, and P2P process owners cannot easily determine which variants are common, which cases experience abnormal duration or repeated processing, where reversals and changes are concentrated, or which transactions deserve investigation first.

This project will transform the event log into controlled event-level and case-level analytical structures and use transparent business rules to quantify process variants, cycle times, PO changes, invoice/GR reversals, payment-block handling, repeated activities, resource handoffs, and exception concentration across relevant business dimensions.

4. Business Rationale

- Manual case-by-case investigation can hide recurring process patterns.
- PO changes, reversals, and repeated processing may increase operational effort and lead time.
- Long-tail cycle times can be hidden by simple averages.
- Raw exception counts can unfairly penalize high-volume vendors unless rates and denominators are considered.
- Without a defined case-prioritization method, operations may spend effort on low-impact cases.
- Inconsistent KPI definitions reduce trust in analytics and make cross-team discussion harder.

5. Project Objectives

- Reconstruct actual P2P transaction sequences at case level.
- Identify common, uncommon, and long-running process variants.
- Measure end-to-end cycle time using robust statistics such as median and P90.
- Quantify PO modifications, invoice reversals, goods-receipt reversals, and payment-block handling.
- Measure repeated activities and resource handoffs as process-complexity indicators.
- Compare exception and cycle-time patterns across vendors and relevant purchasing dimensions.
- Create transparent investigation-priority criteria for case review.
- Deliver evidence-based process-improvement recommendations without overstating causation or financial impact.

6. Scope

6.1 In Scope

- Event-log profiling and validation.
- Event-level and case-level analytical models.
- Process variant analysis.
- Cycle-time analysis.
- PO change/rework analysis.
- Invoice and goods-receipt reversal analysis.
- Payment-block handling analysis.
- Repeated activity and resource-handoff analysis.
- Vendor and purchasing-dimension benchmarking.
- Case investigation prioritization.
- SQL, Python, Excel, Power BI, and BA deliverables.

6.2 Out of Scope

- Fraud detection or fraud prediction.
- Supplier credit-risk or supplier-quality scoring.
- Employee performance evaluation.
- Unsupported financial-loss, savings, or ROI claims.
- Contract price validation without contract/price data.
- Due-date compliance without due-date and payment-term data.
- Automated ERP write-back or production workflow automation.

7. Stakeholders

Stakeholder	Primary Need	Role in Project
Finance / P2P Sponsor	Executive visibility and prioritization	Project direction / executive decisions
Head of Procurement	PO rework and supplier/category process performance	Validate procurement rules and actions
Accounts Payable Manager	Invoice reversals, block handling, clearing	Validate AP exception logic
P2P Process Owner	Variants, bottlenecks, standardization	Approve process definitions and priorities
Internal Audit / Compliance	Traceable logic and evidence	Review control-sensitive interpretation
Procurement / AP Operations	Case-level investigation context	Validate operational meaning
Data / BI Analyst	Correct grain, KPIs, model	Build and validate analytics
ERP / Source-System Support	Event semantics	Clarify ambiguous system events

8. Key Business Questions

1. What process variants occur and which ones dominate the case population?
2. Which variants and segments have the longest median and P90 cycle times?
3. How frequently are purchase orders modified and where is modification concentrated?
4. How frequently are invoice receipts reversed?
5. How frequently are goods receipts reversed?
6. Which cases contain payment-block handling and how do they progress?
7. Which vendors have disproportionately high exception rates after controlling for transaction volume?
8. Are repeated activities and resource handoffs associated with longer processing time?
9. How do 2-way and 3-way matching process groups differ in cycle time and exception behavior?
10. Which cases combine multiple investigation signals and should be reviewed first?

9. Preliminary KPI Framework

These KPIs are candidates. Final formulas will be frozen only after data-understanding and business-rule validation.

Candidate KPI	Business Meaning
Total Cases	Distinct P2P cases, not event rows
Median Cycle Time	Median case end-to-end duration
P90 Cycle Time	90th percentile case duration
Number of Process Variants	Distinct ordered activity sequences
Top Variant Coverage	Share of cases covered by the most frequent variant(s)
PO Change Rate	Eligible cases containing validated PO-change events / eligible cases
Invoice Reversal Rate	Cases containing Cancel Invoice Receipt / eligible cases
GR Reversal Rate	Cases containing Cancel Goods Receipt / eligible cases
Payment-Block Handling Rate	Cases containing set/remove block event using documented denominator
Repeated Activity Case Rate	Cases containing repeated activities / eligible cases
Median Resource Handoffs	Median consecutive resource changes per case

10. Assumptions, Constraints, and Analytical Boundaries

- The event log is anonymized; the stakeholder and management scenario is a portfolio construct aligned to the data.
- Event timestamps will be validated before cycle-time metrics are finalized.
- Repeated activity is not automatically classified as rework.
- A rare variant is not automatically a compliance failure.
- Payment-block events will be interpreted cautiously because set/remove patterns may not be symmetric.
- Monetary values will not be aggregated at event grain until their case-level semantics are validated.
- Associations found in the event data will not be described as proven causation without supporting evidence.
- An exception is a signal for investigation, not proof of fraud, loss, supplier failure, or employee underperformance.

11. Project Success Measures

The analytical solution will be considered successful when it enables stakeholders to:

- Reconstruct the actual Purchase-to-Pay process from event-level data.
- Measure process cycle time at case level using clearly defined calculation rules.
- Identify and quantify purchase-order modifications, invoice reversals, goods-receipt reversals, and payment-block handling.
- Compare exception rates across relevant vendors, spend areas, document types, and P2P process categories.
- Identify common and unusual process variants and surface long-running cases for investigation.
- Provide transaction-level drill-through so users can review the exact event history behind an analytical signal.
- Distinguish event-level metrics from case-level KPIs to prevent incorrect aggregation.
- Maintain traceability between business questions, analytical rules, KPIs, and source data.
- Highlight high-complexity or unusual cases without automatically classifying them as fraud, non-compliance, or financial loss.
- Provide evidence-based insights that support process investigation, prioritization, and improvement decisions.

Decision principle: The project treats analytical exceptions as signals for investigation. Recommendations will be based on traceable evidence and will not overstate causation, financial impact, or control failure.